

Rafi Zaman

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Electrical & Computer Engineering + Machine Learning

Expected 2028

- Relevant Coursework: LLM Systems (PhD), Parallel Computing (PhD), Logic Design & Verification, Distributed Systems, Structure & Design of Digital Systems, Data Structures & Algorithms, Object-Oriented Programming, Concepts of Mathematics, Computer Systems, Analog Devices, Mathematics of Electrical Engineering

PROFESSIONAL/ACADEMIC EXPERIENCE

Machine Learning Engineer Intern

Austin, TX

IBM

April 2026 – Present

- Incoming ML Engineer on the WatsonX Orchestrate/AI Agents team

Machine Learning Infrastructure Intern

New York, NY

DeepSpace

January 2026 – April 2026

- Built AgentUsageTracker to monitor per-command LLM cost attribution across the agent pipeline tracking input/output token usage, applying model pricing, and logging structured billing events via API POST
- Built ML pipeline for ASL sign language recognition trained on 1000+ database entries; Developed Miyagi TTS API to generate free, accessible educational video content for users
- Developed D3.js visual rendering engine with neural network-based layout algorithms processing 50+ diagram types to generate dynamic SVG visualizations (neural networks, molecular structures, code flows) with sub-200ms render time

Firmware Engineering Intern

Pittsburgh, PA

Carnegie Mellon University Biorobotics

April 2025 – October 2025

- Led ESP32 embedded systems for TeleLine ventilator monitoring with real-time 3-point anomaly detection algorithm identifying 75% flow/temperature deviations and achieving 100ms HTTP POST response times
- Designed PSoC hardware schematic with dual UART channels and GPIO mapping; implemented embedded C firmware with UART data parsing, sliding window detection for 10 SLM flow/2°C temperature anomalies

PROJECTS

Winner, BNY Mellon Track at TartanHacks | *Python, Black-Scholes, FinBERT*

February 2026

- Developed a Python options trading pipeline at CMU's largest hackathon (>1050 participants) that identifies mispriced opportunities by combining Black-Scholes theoretical pricing with sentiment analysis using FinBERT and VADER on real-time news from Yahoo Finance and NewsAPI.ai

Flash Attention | *PyTorch, Triton, CUDA, Python*

April 2026

- Implemented Flash Attention v2-style forward/backward GPU kernels in Triton with autotuning, avoiding full $N \times N$ attention score materialization to reduce memory usage vs. naive attention
- Built paged KV cache, grouped-query attention (GQA) with $4 \times$ smaller KV memory vs. standard multi-head attention, and prefill/decode wrappers; benchmarked decode at $\sim 28 \times$ faster than prefill at long context

ML-Driven Hardware Control System | *Python, MediaPipe, STM32, PCB Design*

November 2025

- Built real-time hand tracking system using MediaPipe for landmark detection and trained binary classifiers on joint angles and geometric features to extract finger states, applying moving average filtering and hysteresis
- Designed custom PCB with STM32F103, 5-channel PWM LED drivers, and regulated power supply; implemented binary USB-UART protocol with 9-byte packets and checksums for deterministic ML-to-hardware execution

Custom RISC-V Processor Extension (Open Source) | *SystemVerilog, FPGA*

October 2025

- Designed custom 64-bit arithmetic instruction by extending processor datapath for 4-register reads, modifying ALU for carry propagation, and implementing multi-cycle FSM controller for instruction fetch and arithmetic
- Synthesized design onto FPGA with memory-mapped I/O and benchmarked performance with assembly code

LEADERSHIP EXPERIENCE

President, Engineering Honor Society

Potomac, MD

- Directed 80+ member engineering honor society, organizing meetings and hands-on engineering workshops

Vice President, MSA

Potomac, MD

- Established high school's first dedicated prayer room after navigating administrative approvals securing support from faculty and students; Raised \$3000+ to support local mosque initiatives and organized cultural events

TECHNICAL SKILLS

Technical Skills: Python, SQL, R, Java, C, JavaScript, APIs, ML/CV, PyTorch, CUDA, Git, Linux, Open Source

Activities & Interests: Soccer, Carnegie Mellon Racing, Phi Delta Theta, Football, Creative Writing, Cooking